

**This is CS50**

# This is CS50

- No break today, cupcakes at end! 
- Do **say bye or ask questions** during break or after class!
  - What do today's binary bulbs spell?
- The CS50 Hackathon is this Thursday eve!
  - Register at **[cs50.ly/hackathon](https://cs50.ly/hackathon)**. Roommates and friends welcome to register too.

(Took out winter decorations a bit early.)

**This is CS50**

Happy Birthday to Alexis! 🎉

**This is CS50**



GETTING INFORMATION  
FROM **NOT**  
OLIVE TRYING TO  
GET A DRINK  
FROM A  
FIRE HOSE...



what ultimately matters in this course is not so much where  
you end up relative to your classmates but where  
you end up relative to yourself when you began



```
/*
 * hello.c
 *
 * Assignment: Assignment 1
 *
 * Name: David Malan
 *
 * A program to print "Hello, CS50!" on the screen.
 */

#include <stdio.h>

/*
 * main
 */

void main ()
{
    printf ("Hello, CS50!\n");

    exit (0);
}

/*
 * end of hello.c
 */
```

(-2) for hello.out, we wanted output of hello,  
not of make.

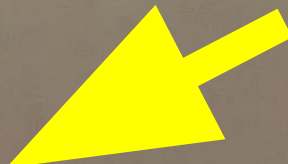
```
/*
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 *
 * Assignment: Assignment 1
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 * Name: David Malan
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#include <stdio.h>

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 * main
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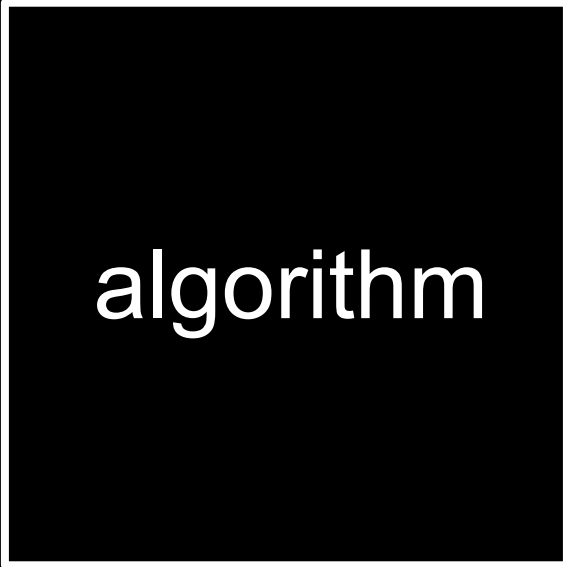
void main ()
{
    printf ("Hello, CS50!\n");
    exit (0);
}

/*
 * end of hello.c
 */
```



(-2) for hello.out, we wanted output of hello,  
not of make.





algorithm

Algorithm - Precise sequence of steps  
for getting something done.

Imp't: Precision!  
Correctness!

Programming: Process of taking an algorithm  
and putting it into a language  
a computer can process.

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# CS50 Pictionary



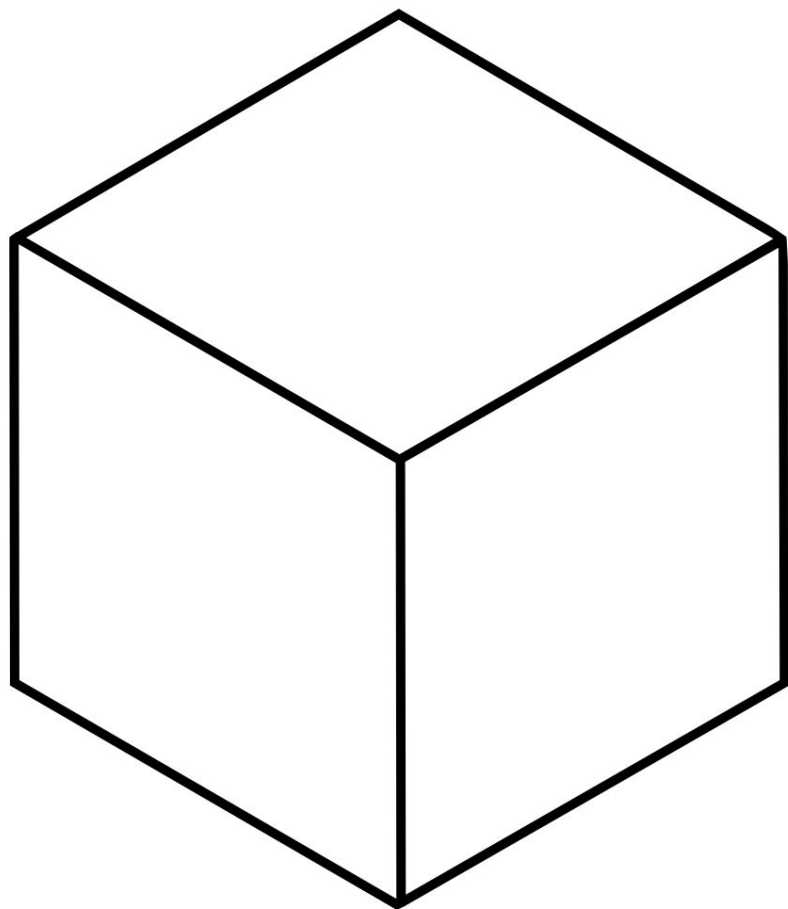


















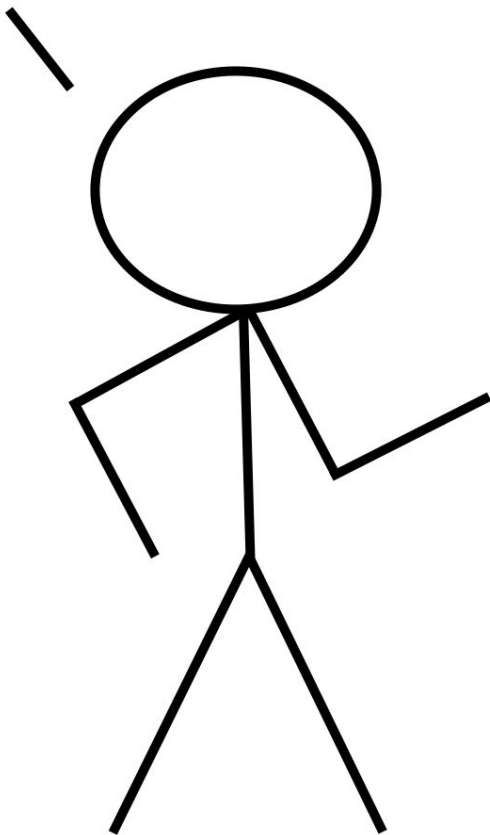








Hi



Week 0 Scratch

Week 1 C

Week 2 Arrays

Week 3 Algorithms

Week 4 Memory

Week 5 Data Structures

Week 6 Python

Week 7 SQL, Artificial Intelligence

Week 8 HTML, CSS, JavaScript

Week 9 Flask

Week 10 The End



# Final Project

all that we ask is that you build something of interest to  
you, that you solve an actual problem, that you impact  
campus, or that you change the world

strive to create something that outlives this course

# CS50 Hackathon

















THIS IS CS50

PRESENT  
10-11-12  
shuttle.harvard.edu

COLLECTING CASH

HARVARD  
SHUTTLE

36442



HACKATHON

The image features a complex, golden Art Deco style background. It consists of a dense network of geometric lines, including squares, rectangles, and triangles, some of which are filled with parallel lines. The overall design is symmetrical and highly detailed. The word 'HACKATHON' is prominently displayed in the center, rendered in a bold, stylized font with vertical hatching. Below the text is a circular emblem containing a stylized 'G' or similar symbol. The entire composition is set against a dark, possibly black, background, which makes the golden elements stand out.

CS50 Fair





















I took CS50.

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# CS50 Charades 🎭

# Recursion

# Python

Duck

# Boolean

# Array



# Loop

# Pointer

# Conditional

# Overflow

# Memory

# Emoji

Tree

# Phonebook



Mario

Big O

Cash

# Keyboard

# Lecture

Credit

# Search

DNA



Sort

Image

# Hexadecimal

Variable

Bug

# Scratch

Hello, world

# Debug



# String

# Integer

Error

RGB

# Dictionary

# Float

# Visual Studio Code for CS50

cs50.dev

# Visual Studio Code

[code.visualstudio.com](https://code.visualstudio.com)



# Visual Studio Code

`code.visualstudio.com`

`cs50.readthedocs.io/cs50.dev`

# Install an editor

<https://code.visualstudio.com/>

<https://cursor.com/>

<https://windsurf.com/>

...

# Install command-line tools

<https://developer.apple.com/xcode/>

<https://learn.microsoft.com/en-us/windows/wsl/about>

...

# Learn Git

[https://youtu.be/MJUJ4wbFm\\_A](https://youtu.be/MJUJ4wbFm_A)

...

# Host a website

<https://docs.github.com/pages>

<https://www.netlify.com/>

...

# Host a web app

<https://aws.amazon.com/education/awseducate/>

<https://azure.microsoft.com/free/students/>

<https://cloud.google.com/edu/students>

<https://www.cloudflare.com/students/>

<https://education.github.com/pack>

<https://www.heroku.com/>

<https://vercel.com/>

...

# Ask AI

<https://chatgpt.com/>

<https://claude.ai/>

<https://gemini.google.com/>

<https://github.com/copilot>

<https://openai.com/codex/>

<https://v0.app/>

...

# Ask questions

<https://www.reddit.com/r/learnprogramming/>

<https://www.reddit.com/r/programming/>

<https://stackoverflow.com/>

<https://serverfault.com/>

...



# Read news

<https://techcrunch.com/>

<https://news.ycombinator.com/>

...

# Take Classes

[edx.org/cs50](https://edx.org/cs50)

Apply

# Stay in touch

[cs50.harvard.edu/communities](https://cs50.harvard.edu/communities)

Thank you



Office for the Arts at Harvard  
**Memorial Hall / Lowell Hall Complex**



**Education Support Services**  
Faculty of Arts and Sciences  
Harvard University



THE RESTAURANT  
CHANGSHO



CS50





# CS50 Quiz Show 🏆



# CS50 Quiz Show 🏆

**This was CS50**





**This was CS50**